



1. Description

1.1. Project

Project Name	UMHController
Board Name	custom
Generated with:	STM32CubeMX 6.11.1
Date	06/29/2026

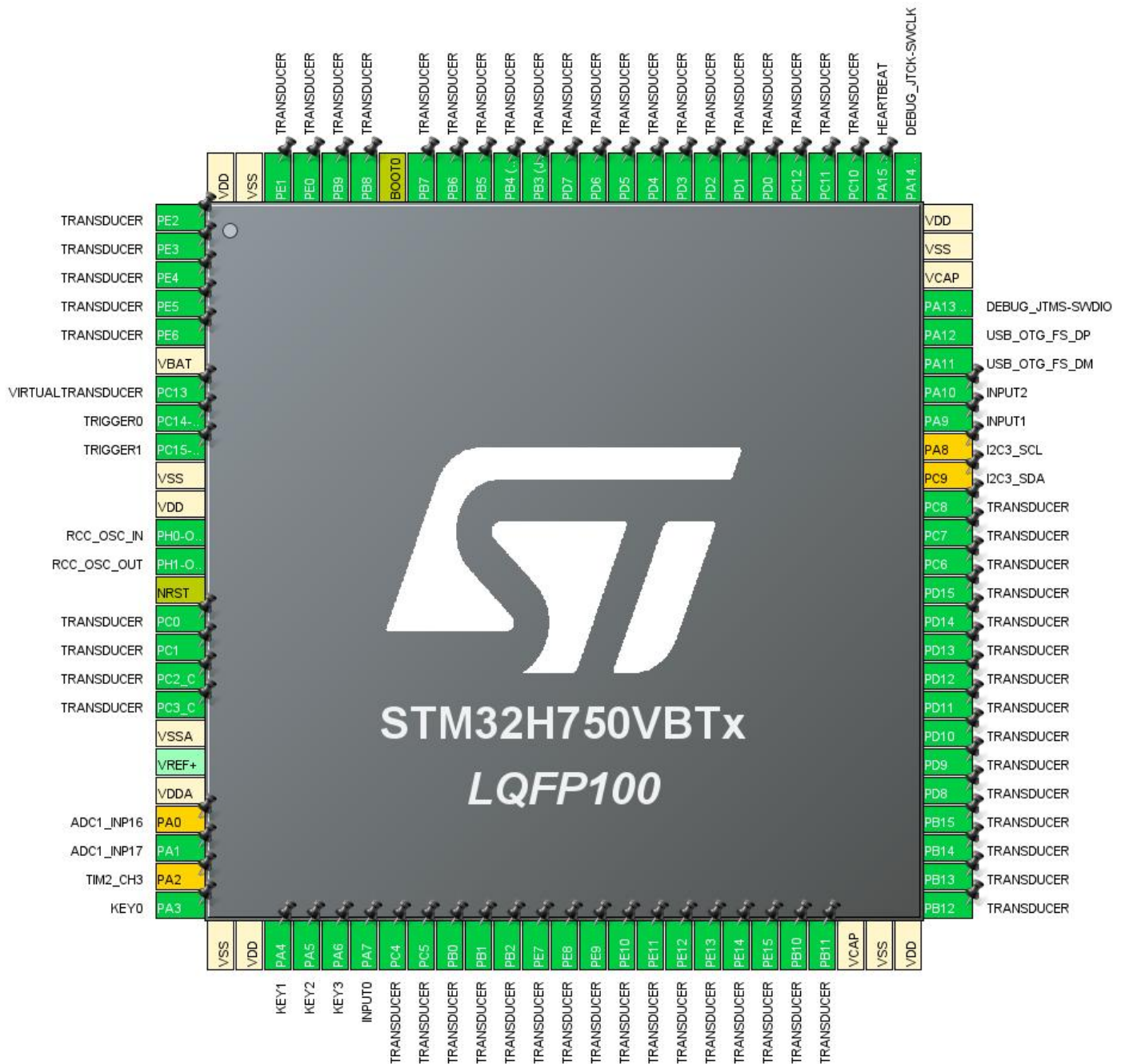
1.2. MCU

MCU Series	STM32H7
MCU Line	STM32H750 Value line
MCU name	STM32H750VBTx
MCU Package	LQFP100
MCU Pin number	100

1.3. Core(s) information

Core(s)	ARM Cortex-M7
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2 *	I/O	GPIO_Output	TRANSDUCER
2	PE3 *	I/O	GPIO_Output	TRANSDUCER
3	PE4 *	I/O	GPIO_Output	TRANSDUCER
4	PE5 *	I/O	GPIO_Output	TRANSDUCER
5	PE6 *	I/O	GPIO_Output	TRANSDUCER
6	VBAT	Power		
7	PC13 *	I/O	GPIO_Output	VIRTUALTRANSDUCER
8	PC14-OSC32_IN (OSC32_IN) *	I/O	GPIO_Output	TRIGGER0
9	PC15-OSC32_OUT (OSC32_OUT) *	I/O	GPIO_Output	TRIGGER1
10	VSS	Power		
11	VDD	Power		
12	PH0-OSC_IN (PH0)	I/O	RCC_OSC_IN	
13	PH1-OSC_OUT (PH1)	I/O	RCC_OSC_OUT	
14	NRST	Reset		
15	PC0 *	I/O	GPIO_Output	TRANSDUCER
16	PC1 *	I/O	GPIO_Output	TRANSDUCER
17	PC2_C *	I/O	GPIO_Output	TRANSDUCER
18	PC3_C *	I/O	GPIO_Output	TRANSDUCER
19	VSSA	Power		
21	VDDA	Power		
22	PA0 **	I/O	ADC1_INP16	
23	PA1	I/O	ADC1_INP17	
24	PA2 **	I/O	TIM2_CH3	
25	PA3 *	I/O	GPIO_Input	KEY0
26	VSS	Power		
27	VDD	Power		
28	PA4 *	I/O	GPIO_Input	KEY1
29	PA5 *	I/O	GPIO_Input	KEY2
30	PA6 *	I/O	GPIO_Input	KEY3
31	PA7 *	I/O	GPIO_Input	INPUT0
32	PC4 *	I/O	GPIO_Output	TRANSDUCER
33	PC5 *	I/O	GPIO_Output	TRANSDUCER
34	PB0 *	I/O	GPIO_Output	TRANSDUCER
35	PB1 *	I/O	GPIO_Output	TRANSDUCER

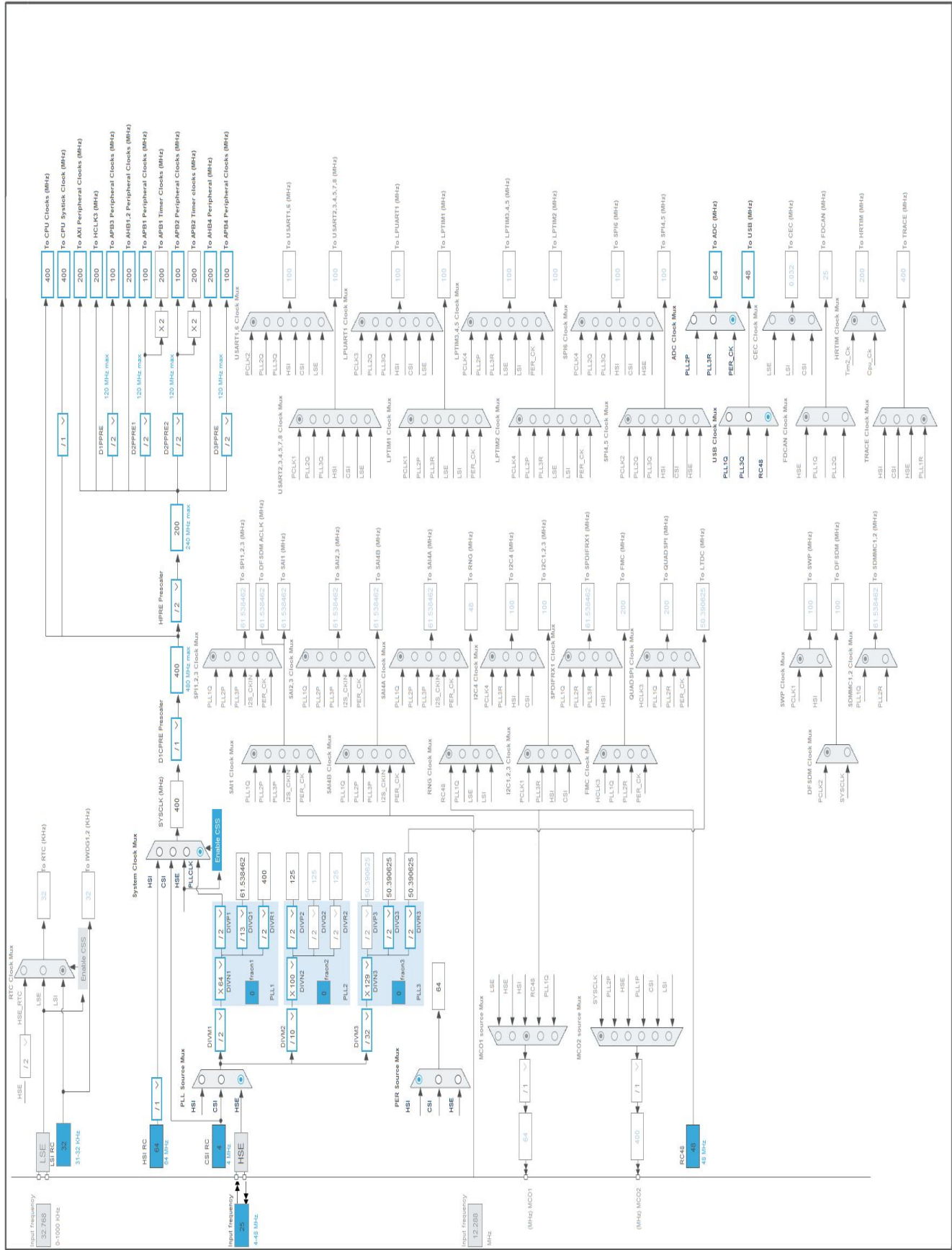
Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
36	PB2 *	I/O	GPIO_Output	TRANSDUCER
37	PE7 *	I/O	GPIO_Output	TRANSDUCER
38	PE8 *	I/O	GPIO_Output	TRANSDUCER
39	PE9 *	I/O	GPIO_Output	TRANSDUCER
40	PE10 *	I/O	GPIO_Output	TRANSDUCER
41	PE11 *	I/O	GPIO_Output	TRANSDUCER
42	PE12 *	I/O	GPIO_Output	TRANSDUCER
43	PE13 *	I/O	GPIO_Output	TRANSDUCER
44	PE14 *	I/O	GPIO_Output	TRANSDUCER
45	PE15 *	I/O	GPIO_Output	TRANSDUCER
46	PB10 *	I/O	GPIO_Output	TRANSDUCER
47	PB11 *	I/O	GPIO_Output	TRANSDUCER
48	VCAP	Power		
49	VSS	Power		
50	VDD	Power		
51	PB12 *	I/O	GPIO_Output	TRANSDUCER
52	PB13 *	I/O	GPIO_Output	TRANSDUCER
53	PB14 *	I/O	GPIO_Output	TRANSDUCER
54	PB15 *	I/O	GPIO_Output	TRANSDUCER
55	PD8 *	I/O	GPIO_Output	TRANSDUCER
56	PD9 *	I/O	GPIO_Output	TRANSDUCER
57	PD10 *	I/O	GPIO_Output	TRANSDUCER
58	PD11 *	I/O	GPIO_Output	TRANSDUCER
59	PD12 *	I/O	GPIO_Output	TRANSDUCER
60	PD13 *	I/O	GPIO_Output	TRANSDUCER
61	PD14 *	I/O	GPIO_Output	TRANSDUCER
62	PD15 *	I/O	GPIO_Output	TRANSDUCER
63	PC6 *	I/O	GPIO_Output	TRANSDUCER
64	PC7 *	I/O	GPIO_Output	TRANSDUCER
65	PC8 *	I/O	GPIO_Output	TRANSDUCER
66	PC9 **	I/O	I2C3_SDA	
67	PA8 **	I/O	I2C3_SCL	
68	PA9 *	I/O	GPIO_Input	INPUT1
69	PA10 *	I/O	GPIO_Input	INPUT2
70	PA11	I/O	USB_OTG_FS_DM	
71	PA12	I/O	USB_OTG_FS_DP	
72	PA13 (JTMS/SWDIO)	I/O	DEBUG_JTMS-SWDIO	
73	VCAP	Power		
74	VSS	Power		

Pin Number LQFP100	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
75	VDD	Power		
76	PA14 (JTCK/SWCLK)	I/O	DEBUG_JTCK-SWCLK	
77	PA15 (JTDI) *	I/O	GPIO_Output	HEARTBEAT
78	PC10 *	I/O	GPIO_Output	TRANSDUCER
79	PC11 *	I/O	GPIO_Output	TRANSDUCER
80	PC12 *	I/O	GPIO_Output	TRANSDUCER
81	PD0 *	I/O	GPIO_Output	TRANSDUCER
82	PD1 *	I/O	GPIO_Output	TRANSDUCER
83	PD2 *	I/O	GPIO_Output	TRANSDUCER
84	PD3 *	I/O	GPIO_Output	TRANSDUCER
85	PD4 *	I/O	GPIO_Output	TRANSDUCER
86	PD5 *	I/O	GPIO_Output	TRANSDUCER
87	PD6 *	I/O	GPIO_Output	TRANSDUCER
88	PD7 *	I/O	GPIO_Output	TRANSDUCER
89	PB3 (JTDO/TRACESWO) *	I/O	GPIO_Output	TRANSDUCER
90	PB4 (NJTRST) *	I/O	GPIO_Output	TRANSDUCER
91	PB5 *	I/O	GPIO_Output	TRANSDUCER
92	PB6 *	I/O	GPIO_Output	TRANSDUCER
93	PB7 *	I/O	GPIO_Output	TRANSDUCER
94	BOOT0	Boot		
95	PB8 *	I/O	GPIO_Output	TRANSDUCER
96	PB9 *	I/O	GPIO_Output	TRANSDUCER
97	PE0 *	I/O	GPIO_Output	TRANSDUCER
98	PE1 *	I/O	GPIO_Output	TRANSDUCER
99	VSS	Power		
100	VDD	Power		

* The pin is affected with an I/O function

** The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration



5. Software Project

5.1. Project Settings

Name	Value
Project Name	UMHController
Project Folder	D:\Data\OneDrive\Projects\UMH\Software\UMH Controller
Toolchain / IDE	Makefile
Firmware Package Name and Version	STM32Cube FW_H7 V1.11.2
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

5.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

5.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_TIM1_Init	TIM1
4	MX_ADC3_Init	ADC3
5	MX_ADC1_Init	ADC1
6	MX_USB_DEVICE_Init	USB_DEVICE

1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H750 Value line
MCU	STM32H750VBTx
Datasheet	DS12556 Rev6

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

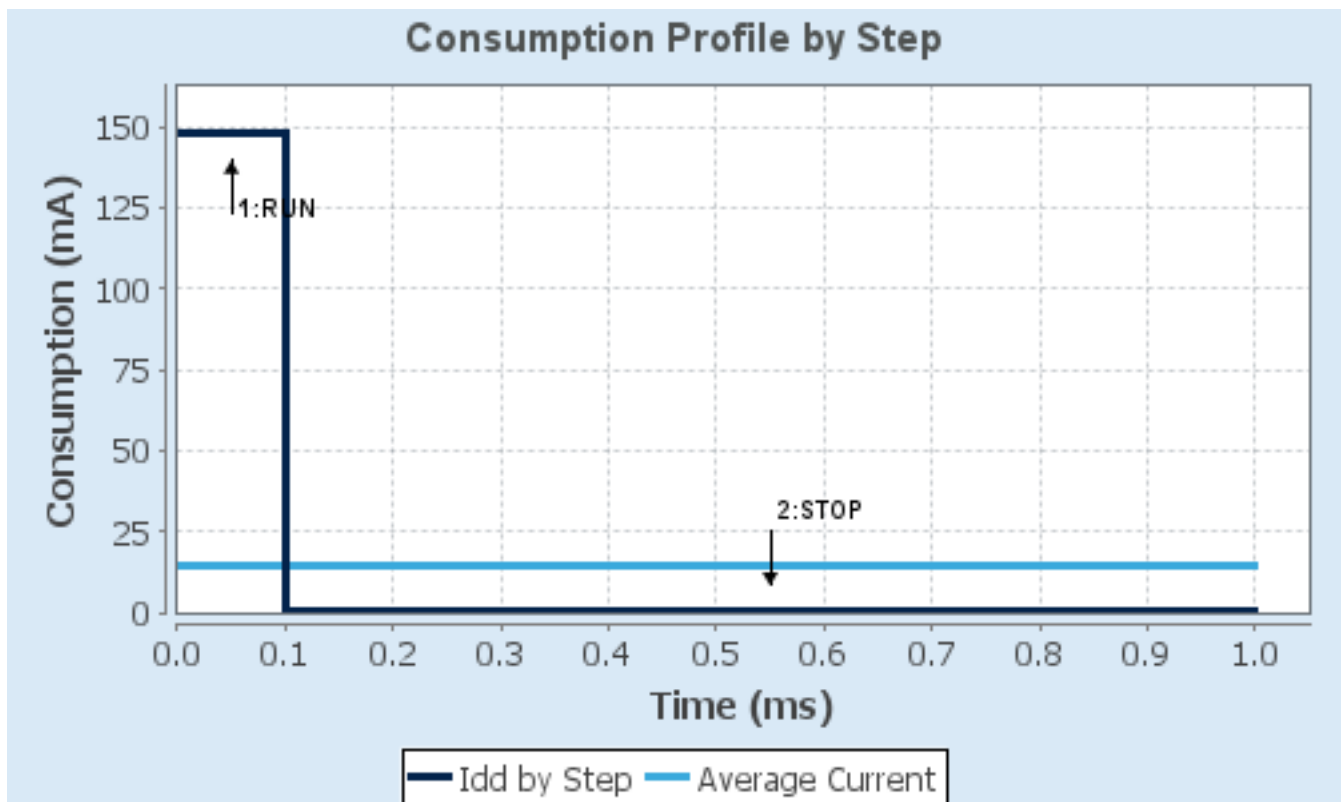
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS0: Scale0-High	SVOS5: System-Scale5
D1 Mode	DRUN/CRUN	DSTANDBY
D2 Mode	DRUN	DSTANDBY
D3 Mode	DRUN	DSTOP
Fetch Type	ITCM	NA
CPU Frequency	480 MHz	0 Hz
Clock Configuration	HSE BYP PLL	Flash-OFF
Clock Source Frequency	24 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	148 mA	150 μ A
Duration	0.1 ms	0.9 ms
DMIPS	1027.0	0.0
Ta Max	105.02	124.98
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	14.94 mA
Battery Life	1 day, 17 hours	Average DMIPS	1027.2001 DMIPS

1.6. Chart



2. Peripherals and Middlewares Configuration

2.1. ADC1

mode: IN17

2.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode Independent mode

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1
Resolution	ADC 16-bit resolution
Scan Conversion Mode	Enabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

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** ADC_Regular_ConversionMode</html>:**

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	2 *
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
<u>Rank</u>	1
Channel	Channel 3 *
Sampling Time	810.5 Cycles *
Offset Number	No offset
Offset Signed Saturation	Disable
<u>Rank</u>	2 *
Channel	Channel 15 *
Sampling Time	810.5 Cycles *
Offset Number	No offset
Offset Signed Saturation	Disable

ADC_Injected_ConversionMode:

Enable Injected Conversions Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode false

2.2. ADC3

mode: Temperature Sensor Channel

mode: Vrefint Channel

2.2.1. Parameter Settings:

ADC_Settings:

Clock Prescaler	Asynchronous clock mode divided by 1
Resolution	ADC 16-bit resolution
Scan Conversion Mode	Enabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
End Of Conversion Selection	End of single conversion
Overrun behaviour	Overrun data preserved
Left Bit Shift	No bit shift
Conversion Data Management Mode	Regular Conversion data stored in DR register only
Low Power Auto Wait	Disabled

ADC_Regular_ConversionMode:

Enable Regular Conversions	Enable
Enable Regular Oversampling	Disable
Number Of Conversion	2 *
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
<u>Rank</u>	1
Channel	Channel Temperature Sensor
Sampling Time	810.5 Cycles *
Offset Number	No offset
Offset Signed Saturation	Disable
<u>Rank</u>	2 *
Channel	Channel Vrefint *
Sampling Time	810.5 Cycles *
Offset Number	No offset
Offset Signed Saturation	Disable

ADC_Injected_ConversionMode:

Enable Injected Conversions Disable

Analog Watchdog 1:

Enable Analog WatchDog1 Mode false

Analog Watchdog 2:

Enable Analog WatchDog2 Mode false

Analog Watchdog 3:

Enable Analog WatchDog3 Mode false

2.3. CORTEX_M7

2.3.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode Disabled

Cortex Interface Settings:

CPU ICache Disabled

CPU DCache Disabled

Cortex Memory Protection Unit Control Settings:

MPU Control Mode **Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers ***

Cortex Memory Protection Unit Region 0 Settings:

MPU Region **Enabled ***

MPU Region Base Address **0x30000000 ***

MPU Region Size **256KB ***

MPU SubRegion Disable **0x0 ***

MPU TEX field level **level 1 ***

MPU Access Permission **ALL ACCESS PERMITTED ***

MPU Instruction Access ENABLE

MPU Shareability Permission DISABLE

MPU Cacheable Permission DISABLE

MPU Bufferable Permission DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 2 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 3 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 4 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 5 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 6 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region Disabled

2.4. DEBUG

Debug: Serial Wire

2.5. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

2.5.1. Parameter Settings:

Power Parameters:

SupplySource	PWR_LDO_SUPPLY
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1

RCC Parameters:

TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

CSI Calibration Value	32
HSI Calibration Value	64

System Parameters:

VDD voltage (V)	3.3
Flash Latency(WS)	2 WS (3 CPU cycle)
Product revision	rev.V

PLL range Parameters:

PLL1 clock Input range	Between 8 and 16 MHz
PLL1 clock Output range	Wide VCO range

2.6. SYS

Timebase Source: SysTick

2.7. TIM1

Clock Source : Internal Clock

2.7.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	99 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	40200 *
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 16 bits value)	0
auto-reload preload	Enable *

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)
Trigger Event Selection TRGO2	Reset (UG bit from TIMx_EGR)

2.8. USB_OTG_FS

Mode: Device_Only

2.8.1. Parameter Settings:

Speed	Full Speed 12MBit/s
Enable internal IP DMA	Disabled
Low power	Disabled
Battery charging	Disabled

Link Power Management	Disabled
Use dedicated end point 1 interrupt	Disabled
VBUS sensing	Disabled
Signal start of frame	Disabled

2.9. USB_DEVICE

Class For FS IP: Communication Device Class (Virtual Port Com)

2.9.1. Parameter Settings:

Basic Parameters:

USBD_MAX_NUM_INTERFACES (Maximum number of supported interfaces)	1
USBD_MAX_NUM_CONFIGURATION (Maximum number of supported configuration)	1
USBD_MAX_STR_DESC_SIZ (Maximum size for the string descriptors)	512
USBD_SELF_POWERED (Enabled self power)	Enabled
USBD_DEBUG_LEVEL (USBD Debug Level)	0: No debug message
USBD_LPM_ENABLED (Link Power Management)	1: Link Power Management supported

Class Parameters:

USB CDC Rx Buffer Size	2048
USB CDC Tx Buffer Size	2048

2.9.2. Device Descriptor:

Device Descriptor:

VID (Vendor Identifier)	1155
LANGID_STRING (Language Identifier)	English(United States)
MANUFACTURER_STRING (Manufacturer Identifier)	STMicroelectronics

Device Descriptor FS:

PID (Product Identifier)	22336
PRODUCT_STRING (Product Identifier)	STM32 Virtual ComPort
CONFIGURATION_STRING (Configuration Identifier)	CDC Config
INTERFACE_STRING (Interface Identifier)	CDC Interface

* User modified value

3. System Configuration

3.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PA1	ADC1_INP17	Analog mode	No pull-up and no pull-down	n/a	
DEBUG	PA13 (JTMS/SWDIO)	DEBUG_JTMS-SWDIO	n/a	n/a	n/a	
	PA14 (JTCK/SWCLK)	DEBUG_JTCK-SWCLK	n/a	n/a	n/a	
RCC	PH0-OSC_IN (PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT (PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
USB_OTG_FS	PA11	USB_OTG_FS_DM	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PA12	USB_OTG_FS_DP	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
Single Mapped Signals	PA0	ADC1_INP16	Analog mode	No pull-up and no pull-down	n/a	
	PA2	TIM2_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PC9	I2C3_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Low	
	PA8	I2C3_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Low	
GPIO	PE2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PC13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	VIRTUALTRANSDUCER
	PC14-OSC32_IN (OSC32_IN)	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TRIGGER0

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PC15-OSC32_OUT	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TRIGGER1
	PC0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PC1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PC2_C	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PC3_C	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PA3	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY0
	PA4	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY1
	PA5	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY2
	PA6	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	KEY3
	PA7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	INPUT0
	PC4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PC5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
					*	
	PE13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PC6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PC7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PC8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PA9	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	INPUT1
	PA10	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	INPUT2
	PA15 (JTDI)	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	HEARTBEAT
	PC10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PC11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PC12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD3	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PD7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB3 (JTDO/TRACESWO)	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB4 (NJTRST)	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High	TRANSDUCER

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
					*	
	PB7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PB9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER
	PE1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Very High *	TRANSDUCER

3.2. DMA configuration

nothing configured in DMA service

3.3. BDMA configuration

nothing configured in DMA service

3.4. MDMA configuration

nothing configured in DMA service

3.5. NVIC configuration

3.5.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
TIM1 update interrupt	true	0	0
USB On The Go FS global interrupt	true	0	0
PVD and AVD interrupts through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1 and ADC2 global interrupts	unused		
TIM1 break interrupt	unused		
TIM1 trigger and commutation interrupts	unused		
TIM1 capture compare interrupt	unused		
FPU global interrupt	unused		
USB On The Go FS End Point 1 Out global interrupt	unused		
USB On The Go FS End Point 1 In global interrupt	unused		
HSEM1 global interrupt	unused		
ADC3 global interrupt	unused		

3.5.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
System tick timer	false	true	true
TIM1 update interrupt	false	true	true
USB On The Go FS global interrupt	false	true	true

*** User modified value**

4. System Views

4.1. Category view

4.1.1. Current

Category view

Power Domain view

Choose filters ...

... by Power Domain

D1

D2

D3

None

Middleware

USB_DEVICE ✓

System Core

Analog

Timers

Connectivity

Multimedia

Security

Computing

Trace and Debug

Power and Thermal

BDMA

CORTEX_M7 ✓

DMA

GPIO ⚠

MDMA

IVIC ✓

RCC ✓

SYS ✓

ADC1 ✖

ADC3 ✓

TIM1 ✓

USB_FS ✓

DEBUG ✓

Page 24

4.1.2. Without filters

Category view

Power Domain view

Choose filters ...

... by Power Domain

D1

D2

D3

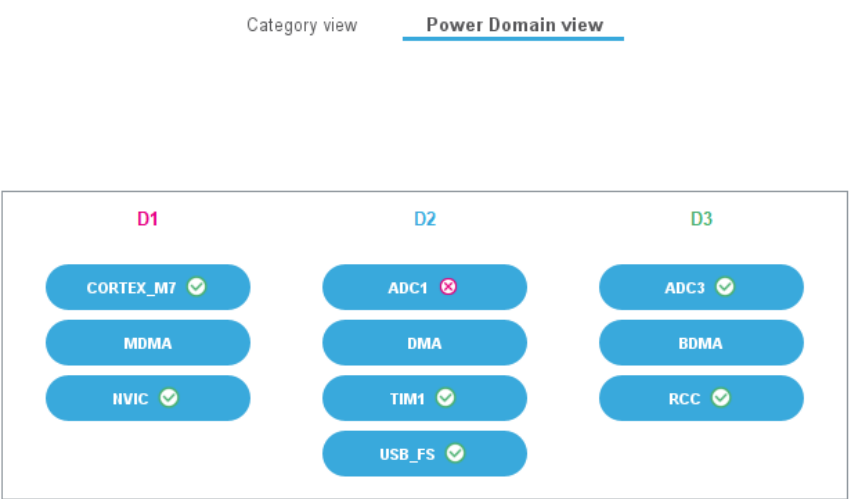
None

Middleware

USB_DEVICE ✓

System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing	Trace and Debug	Power and Thermal
BDMA	ADC1 ✗	TIM1 ✓	USB_FS ✓				DEBUG ✓	
CORTEX_M7 ✓	ADC3 ✓							
DMA								
GPIO ⚠								
MDMA								
IVIC ✓								
RCC ✓								
SYS ✓								

4.2. Power Domain view



5. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32h7_bsd1.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32h7_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32h7-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32h7_series_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32h7rs-lines-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/brstm32h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32f7h7.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7vl.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32h7rs.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
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Application Notes https://www.st.com/resource/en/application_note/an5001-stm32cube-expansion-package-for-stm32h7-series-mdma-stmicroelectronics.pdf

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Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5033-stm32cube-mcu-package-examples-for-stm32h7-series-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbp-d-sink-application-with-stm32cubemx-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4865-lowpower-timer-lptim-applicative-use-cases-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
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Application Notes	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-

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for related Tools lines-and-stm32h7b0-value-line-smart-power-management-expansion-
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for related Tools integrate-the-stl-firmware-into-a-time-critical-user-application-
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Tips beamforming-in-the-pcm-and-pdm-domain-stmicroelectronics.pdf

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Datasheet	https://www.st.com/resource/en/datasheet/dm00496677.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0253-stm32f7-series-and-stm32h7-series-cortexm7-processor-programming-manual-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0433-stm32h742-stm32h743753-and-stm32h750-value-line-advanced-armbased-32bit-mcus-stmicroelectronics.pdf
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Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
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